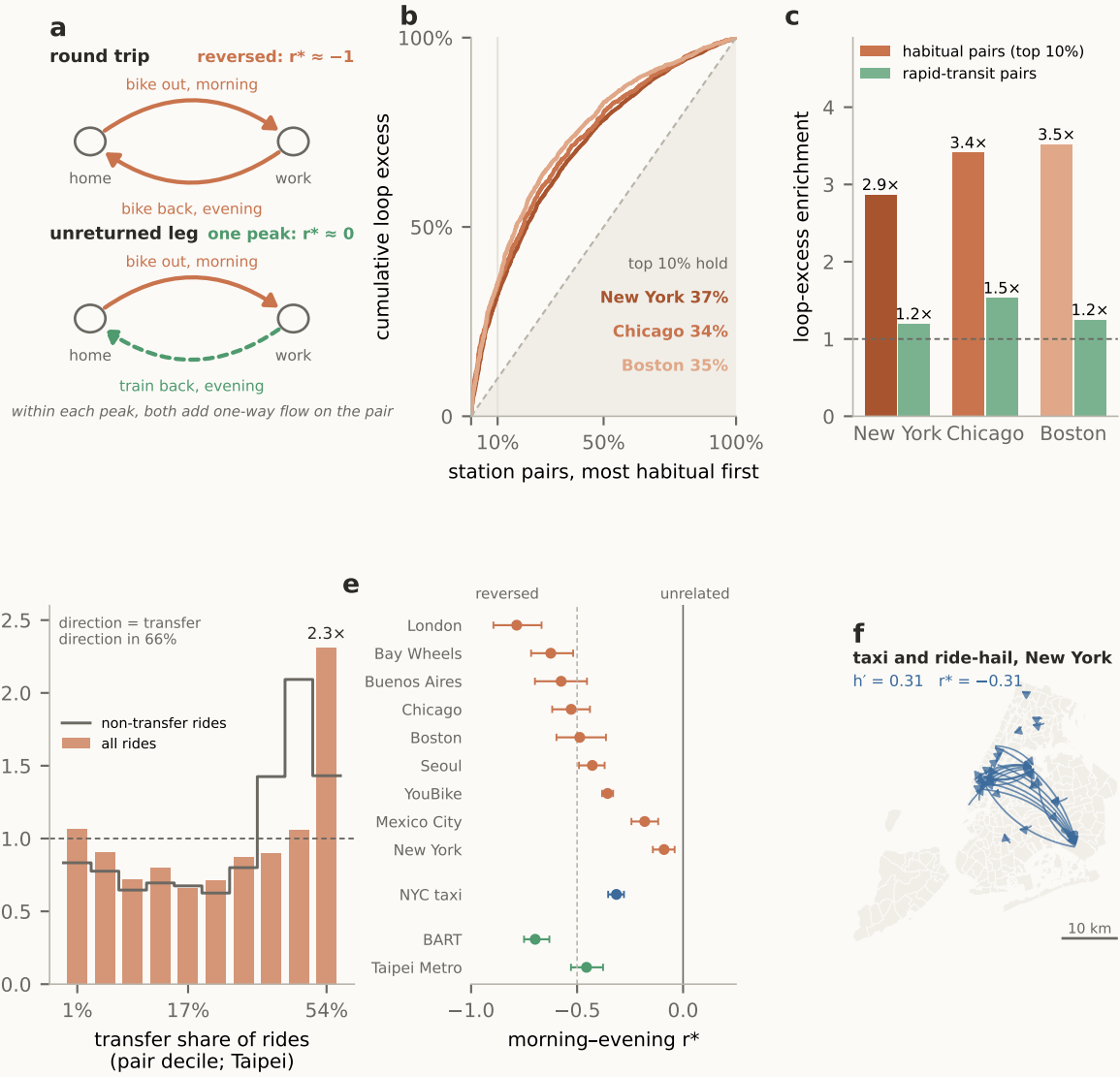

a
round trip
reversed: r* ≈ −1
bike out, morning
home
work
bike back, evening
unreturned leg
one peak: r* ≈ 0
bike out, morning
home
work
train back, evening
within each peak, both add one-way flow on the pair
b
100%
50%
0
cumulative loop excess
top 10% hold
New York 37%
Chicago 34%
Boston 35%
10%
50%
100%
station pairs, most habitual first
c
habitual pairs (top 10%)
rapid-transit pairs
loop-excess enrichment
2.9×
1.2×
3.4×
1.5×
3.5×
1.2×
New York
Chicago
Boston
d
direction = transfer
direction in 66%
2.3×
non-transfer rides
all rides
loop-excess enrichment
1%
17%
54%
transfer share of rides
(pair decile; Taipei)
e
reversed
unrelated
London
Bay Wheels
Buenos Aires
Chicago
Boston
Seoul
YouBike
Mexico City
New York
NYC taxi
BART
Taipei Metro
−1.0
−0.5
0.0
morning–evening r*
f
taxi and ride-hail, New York
h′ = 0.31
r* = −0.31
10 km